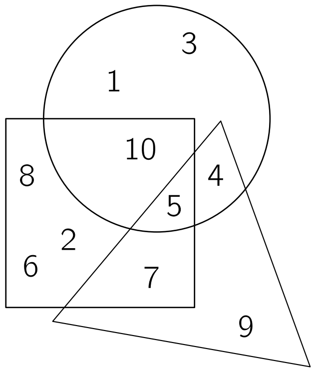
3 points

1. Which number is inside the triangle, the square and the circle?

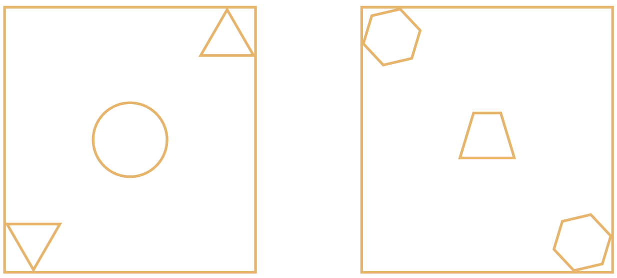
qǐng wèn nǎ ge shù zì tóng shí wèi yú sān jiǎo xíng zhèng fāng xíng hé yuán xíng zhōng
请问哪个数字同时位于三角形、正方形和圆形中?

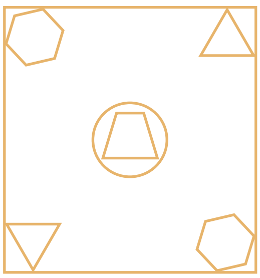


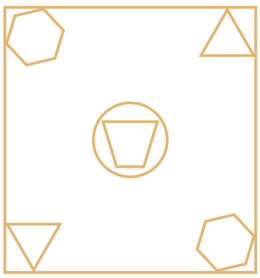
(**A**) 1 (**B**) 4 (**C**) 5 (**D**) 9 (**E**) 12

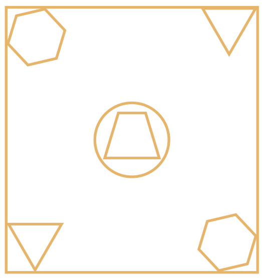
2. Some shapes are printed on 2 pieces of glass. Anna places one on top of the other, without turning either piece. What does she see?

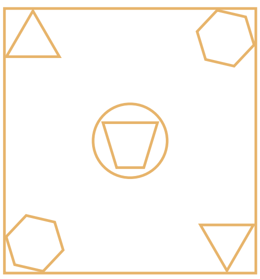
rú tú suǒ shì kuài bō li shàng yìn yǒu yì xiē tú àn xiǎo ān bǎ qí zhōng yí kuài bō li fàng zài lìng yí kuài de shàng miàn rú guǒ bù zhuàn dòng bō li qǐng wèn xiǎo ān huì kàn dào shén me tú àn
如图所示，2块玻璃上印有一些图案。小安把其中一块玻璃放在另一块的上面。如果不转动玻璃，请问小安会看到什么图案?

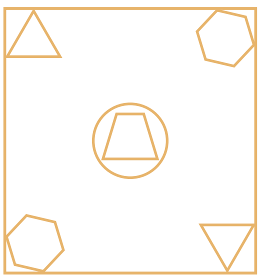


(**A**)


(**B**)


(**C**)


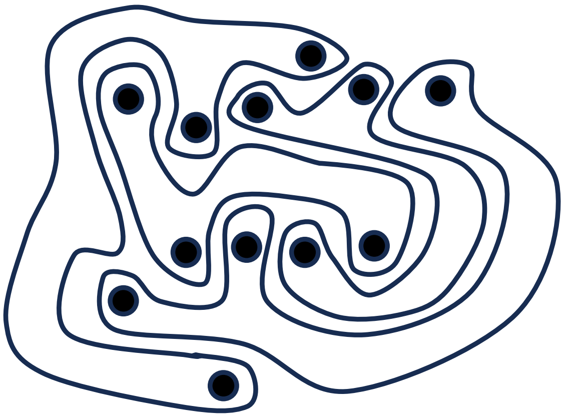
(**D**)


(**E**)


3. The picture shows 4 strange shapes. How many shapes have 3 dots inside?

tú zhōng yǒu gè qí guài de tú xíng, qǐng wèn yǒu jǐ gè tú xíng lǐ miàn bāo hán gè diǎn

图中有 4 个奇怪的图形，请问有几个图形里面包含 3 个点?

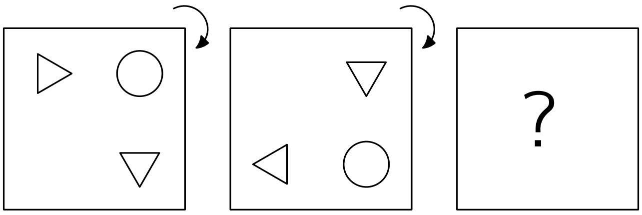


(**A**) 0 (**B**) 1 (**C**) 2 (**D**) 3 (**E**) 4

4. Kevin the kangaroo puts a picture on the table.

dài shǔ xiǎo kǎi bǎ yì fú huà fàng zài zhuō zi shàng

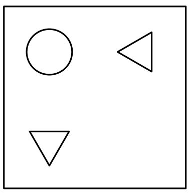
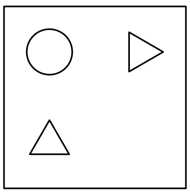
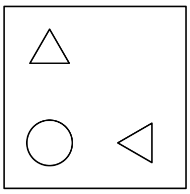
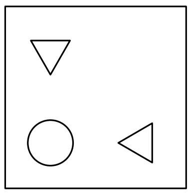
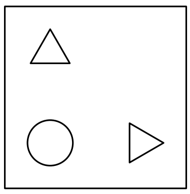
袋鼠小凯把一幅画放在桌子上。



He rotates the picture through a quarter turn, as shown. He then does the same rotation again. What does Kevin see now?

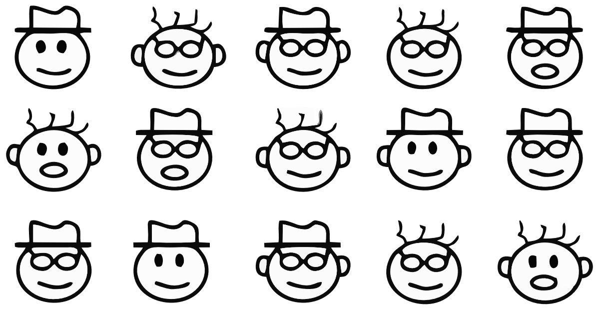
rú tú suǒ shì, tā jiāng zhè fú huà xuán zhuǎn le sì fēn zhī yī quān。rán hòu tā yòu zuò le yí cì tóng yàng de xuán zhuǎn。qǐng wèn xiǎo kǎi xiàn zài kàn dào de shì xià liè nǎ ge xuǎn xiàng?

如图所示，他将这幅画旋转了四分之一圈。然后他又做了一次同样的旋转。请问小凯现在看到的是下列哪个选项?

(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

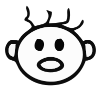
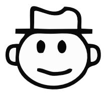
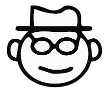
5. In the picture, there are 8 different faces.

tú zhōng yǒu zhāng bù tóng de liǎn
图中有 8 张不同的脸。



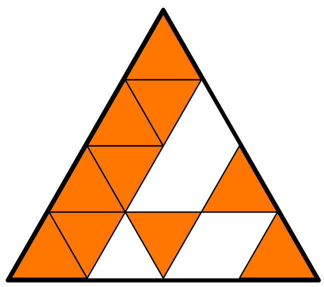
Each face appears twice, except for one. Which face appears only once?

chú le qí zhōng yì zhāng liǎn qí yú de liǎn dōu gè chū xiàn le liǎng cì qǐng wèn nǎ zhāng liǎn zhǐ chū xiàn le yí cì
除了其中一张脸，其余的脸都各出现了两次。请问哪张脸只出现了一次?

(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

6. Bruno is making this large triangle using identical small triangular tiles.

xiǎo bù zhèng zài yòng xiāng tóng de xiǎo sān jiǎo xíng cí zhuān pīn yí gè dà sān jiǎo xíng
小布正在用相同的小三角形瓷砖拼一个大三角形。

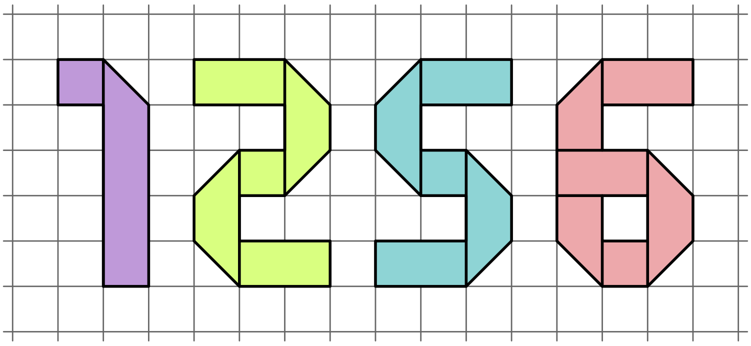


How many more tiles does Bruno need to complete the large triangle?

qǐng wèn xiǎo bù hái xū yào jǐ kuài cí zhuān cái néng pīn chéng zhè gè dà sān jiǎo xíng
请问小布还需要几块瓷砖才能拼成这个大三角形?

(**A**) 3 (**B**) 4 (**C**) 5 (**D**) 6 (**E**) 7

7. Each number below is made using a piece of ribbon.
xià fāng měi gè shù zì dōu shì yòng yì tiáo sī dài zhì zuò ér chéng de
下方每个数字都是用一条丝带制作而成的。

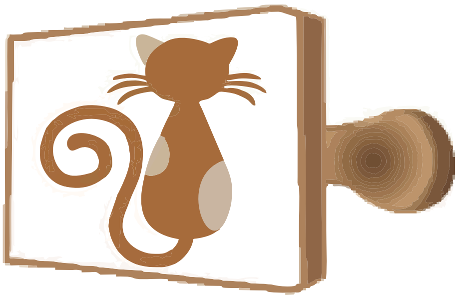


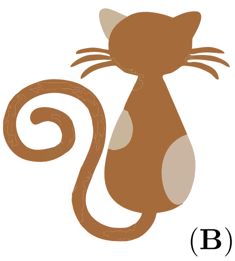
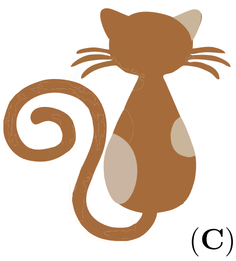
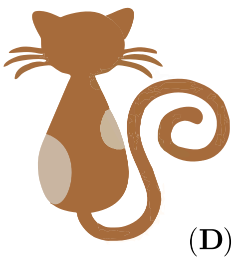
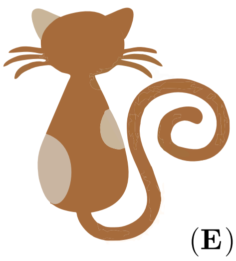
Which piece of ribbon is the longest?
qǐng wèn nǎ tiáo sī dài zuì cháng
请问哪条丝带最长?

(**A**) 1 (**B**) 2 (**C**) 5 (**D**) 6

(**E**) They are all the same length.
tā men de cháng dù dōu yí yàng
它们的长度都一样。

8. Elena uses the stamp shown to make a picture. Which picture does she make?
xiǎo ài yòng xià tú suǒ shì de yìn zhāng gài zhāng, qǐng wèn nǎ ge shì tā dé dào de tú àn
小艾用下图所示的印章盖章，请问哪个是她得到的图案?



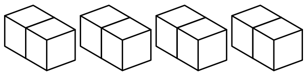
(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

4 points

9. A student has 4 blocks, as shown.

rú tú suǒ shì, yì míng xué shēng yǒu 4 kuài jī mù

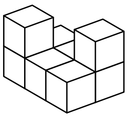
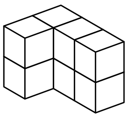
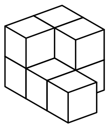
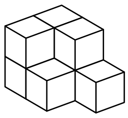
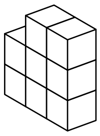
如图所示，一名学生有4块积木。



Which of the following shapes cannot be made using these 4 blocks?

qǐng wèn yòng zhè 4 kuài jī mù bù néng dā chéng xià liè nǎ zhǒng xíng zhuàng

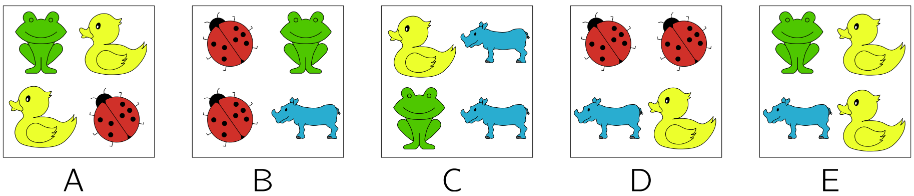
请问用这4块积木不能搭成下列哪种形状?

(A)  **(B)**  **(C)**  **(D)**  **(E)** 

10. Chen has 5 baskets, each containing 4 toys.

xiǎo chén yǒu 5 gè lán zi, měi gè lán zi lǐ yǒu 4 gè wán jù

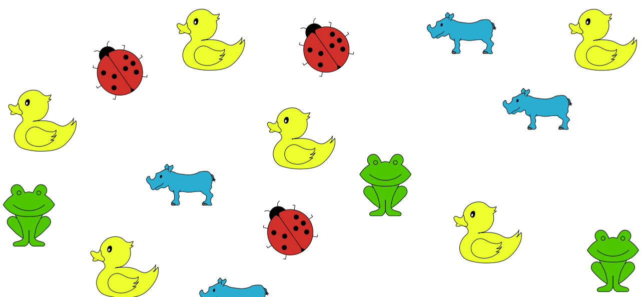
小陈有5个篮子，每个篮子里有4个玩具。



He dropped 4 of the baskets and the toys were mixed up.

tā pèng diào le qí zhōng 4 gè lán zi, lǐ miàn de wán jù yě hùn zài le yì qǐ

他碰掉了其中4个篮子，里面的玩具也混在了一起。



Which basket did he not drop?

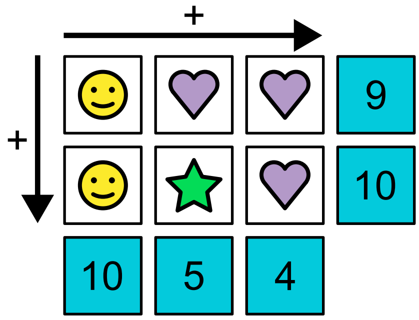
qǐng wèn tā méi yǒu pèng diào nǎ ge lán zi

请问他没有碰掉哪个篮子?

(A) A **(B)** B **(C)** C **(D)** D **(E)** E

11. In the following diagram, each shape represents a different value.

xià tú zhōng měi zhǒng tú xíng dài biǎo yí gè bù tóng de shù zì
下 图 中， 每 种 图 形 代 表 一 个 不 同 的 数 字。



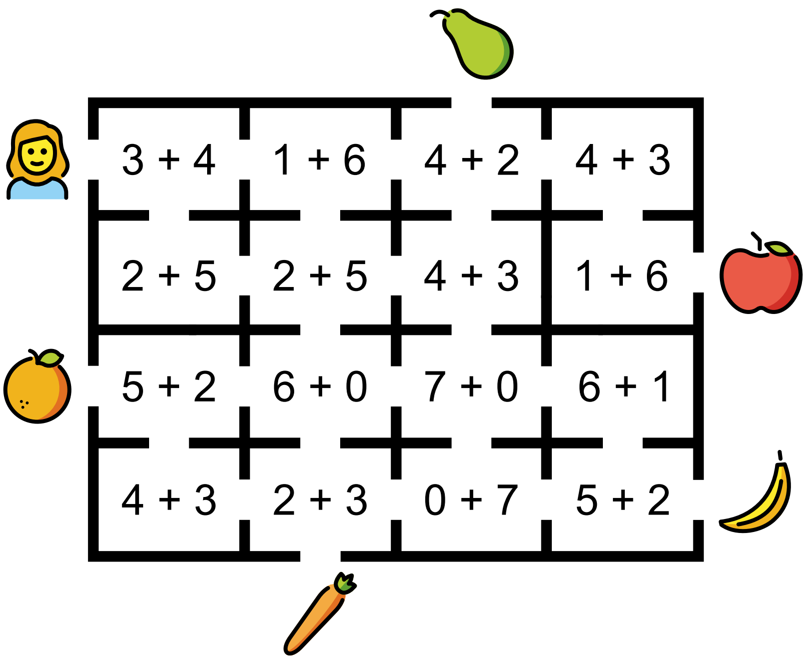
What is the value of ?

qǐng wèn dài biǎo de shù zì shì jǐ
请 问  代 表 的 数 字 是 几?

(A) 2 **(B)** 3 **(C)** 4 **(D)** 5 **(E)** 6

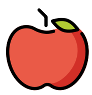
12. Catrina wants to walk through the maze so that she visits only rooms where the answer to the sum is 7.

xiǎo kǎ xiǎng yào chuān guò mí gōng bìng qiě tā zhǐ néng jīng guò hé wéi de fáng jiān
小 卡 想 要 穿 过 迷 宫， 并 且 她 只 能 经 过 和 为 7 的 房 间。



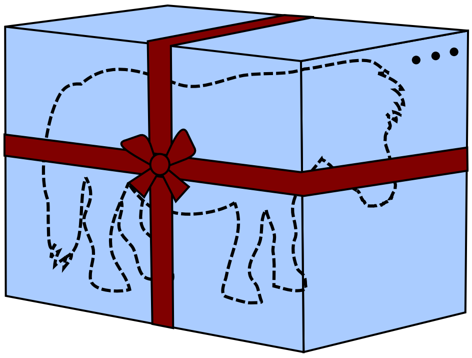
Which object can Catrina reach?

qǐng wèn xiǎo kǎ zuì zhōng kě yǐ huò dé nǎ zhǒng shí wù
请 问 小 卡 最 终 可 以 获 得 哪 种 食 物?

(A)  **(B)**  **(C)**  **(D)**  **(E)** 

13. A toy pony is inside a box that is 1 metre tall, 1 metre wide and 2 metres long.

yì pǐ wán jù xiǎo mǎ zài yí gè gāo mǐ kuān mǐ cháng mǐ de hé zi lǐ
一匹玩具小马在一个高 1 米、宽 1 米、长 2 米的盒子里。



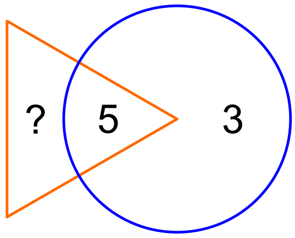
A ribbon goes around the box, as shown. The knot uses an extra 1 metre of ribbon. How long is the ribbon in total?

rú tú suǒ shì hé zi shàng jì zhe yì tiáo sī dài hú dié jié de wèi zhi yòng le mǐ cháng de sī dài qǐng wèn zhè tiáo sī dài de zǒng cháng dù shì duō shǎo
如图所示，盒子上系着一条丝带。蝴蝶结的位置用了 1 米长的丝带。请问这条丝带的总长度是多少?

(**A**) 9 metres 9 米 (mǐ) (**B**) 11 metres 11 米 (mǐ) (**C**) 13 metres 13 米 (mǐ)
(**D**) 15 metres 15 米 (mǐ) (**E**) 17 metres 17 米 (mǐ)

14. The sum of the numbers in the triangle should be twice the sum of the numbers in the circle.

sān jiǎo xíng zhōng de shù zì zhī hé shì yuán zhōng shù zì zhī hé de liǎng bèi
三角形中的数字之和是圆中数字之和的两倍。



What number must replace the question mark?

qǐng wèn tú zhōng wèn hào yí dìng kě yǐ yòng nǎ ge shù zì dài tì
请问图中问号一定可以用哪个数字代替?

(**A**) 3 (**B**) 5 (**C**) 8 (**D**) 11 (**E**) 16

15. A line of pictures is made by repeating this pattern of 5 pictures 😊 👻 🐱 🌙 🔥 always in the same order.

xià fāng yì pái tú àn shì jiāng zhè wǔ gè tú àn 😊 👻 🐱 🌙 🔥 àn zhào tóng
下 方 一 排 图 案 是 将 这 五 个 图 案 😊 👻 🐱 🌙 🔥 按 照 同
yàng de shùn xù bù duàn chóng fù pái liè ér chéng de
样 的 顺 序 不 断 重 复 排 列 而 成 的。

😊 👻 🐱 🌙 🔥 😊 👻 🐱 🌙 🔥 😊 👻 🐱 🌙 🔥 ···

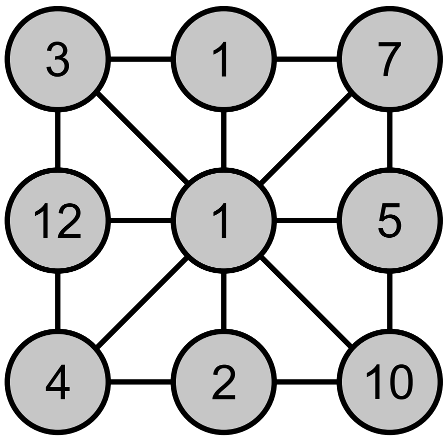
Which picture is in the 27th position in the line?

qǐng wèn dì gè tú àn shì shén me
请 问 第 27 个 图 案 是 什 么?

(**A**) 😊 (**B**) 👻 (**C**) 🐱 (**D**) 🌙 (**E**) 🔥

16. One of the numbers in the picture is equal to the sum of the numbers connected directly to it. Which number is this?

tú zhōng yǒu yí gè shù zì děng yú yǔ tā zhí jiē xiāng lián de shù zì zhī hé, qǐng
图 中 有 一 个 数 字 等 于 与 它 直 接 相 连 的 数 字 之 和， 请
wèn shì nǎ yí gè shù zì
问 是 哪 一 个 数 字?

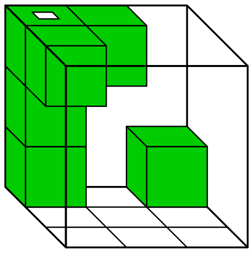


(**A**) 3 (**B**) 5 (**C**) 7 (**D**) 10 (**E**) 12

5 points

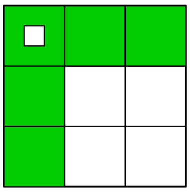
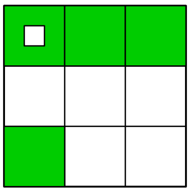
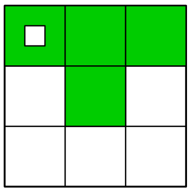
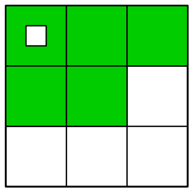
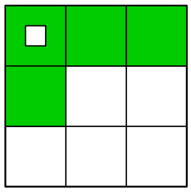
17. Chiara has a transparent box containing 6 small cubes, as shown.

xiǎo qí yǒu yí gè tòu míng de xiāng zi lǐ miàn bāo hán gè xiǎo lì fāng tǐ
小 奇 有 一 个 透 明 的 箱 子， 里 面 包 含 6 个 小 立 方 体，
rú tú suǒ shì
如 图 所 示。



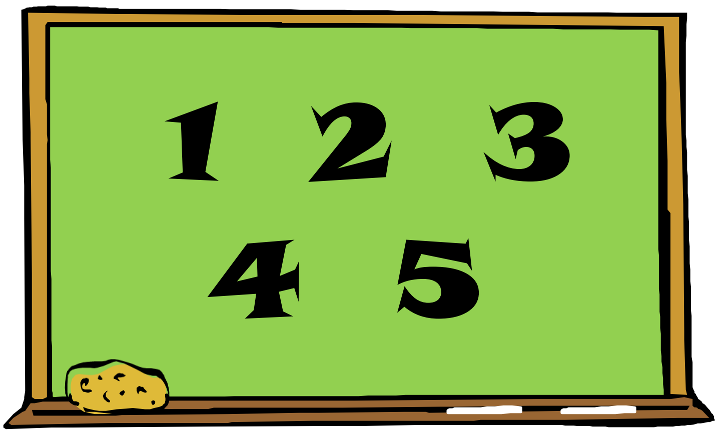
What does Chiara see if she looks at the box from above?

rú guǒ xiǎo qí cóng shàng wǎng xià kàn zhè gè xiāng zi qǐng wèn tā kàn dào de shì xià
如 果 小 奇 从 上 往 下 看 这 个 箱 子， 请 问 她 看 到 的 是 下
liè nǎ zhāng tú
列 哪 张 图?

(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

18. Esteban wants to pick two numbers from the board and add them together.

xiǎo yī xiǎng cóng hēi bǎn shàng xuǎn chū liǎng gè shù zì bìng xiāng jiā
小 伊 想 从 黑 板 上 选 出 两 个 数 字 并 相 加。



How many different results could Esteban get?

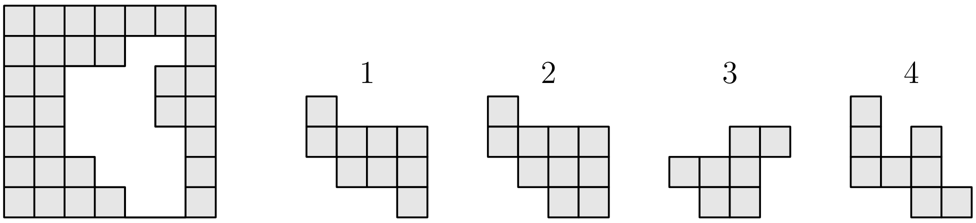
qǐng wèn xiǎo yī kě yǐ dé dào duō shǎo gè bù tóng de jié guǒ
请 问 小 伊 可 以 得 到 多 少 个 不 同 的 结 果?

(**A**) 5 (**B**) 6 (**C**) 7 (**D**) 8 (**E**) 10

19. Which two pieces can be used to complete the grid without overlapping?

zài bù chóng dié de qíng kuàng xià, qǐng wèn yòng nǎ liǎng kuài tú xíng kě yǐ jiāng zhè gè dà fāng gé bǔ chōng wán zhěng?

在不重叠的情况下，请问用哪两块图形可以将这个大方格补充完整?

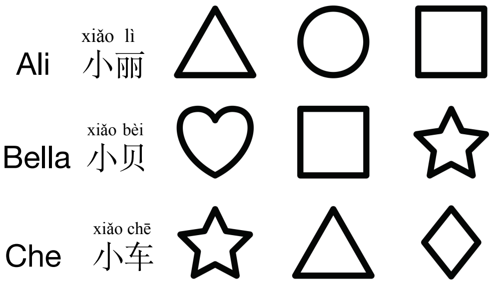


(**A**) 1 and 2 1 和 2 (hé)
(**B**) 1 and 3 1 和 3 (hé)
(**C**) 3 and 4 3 和 4 (hé)
(**D**) 2 and 4 2 和 4 (hé)
(**E**) 2 and 3 2 和 3 (hé)

20. Ali, Bella, Che and Dimitry each have 3 shapes. Each child has exactly one shape in common with every other child.

xiǎo lì、xiǎo bèi、xiǎo chē hé xiǎo dí gè yǒu 3 gè tú xíng, měi wèi xiǎo péng yǒu dōu qià hǎo yǒu yí gè hé qí tā xiǎo péng yǒu xiāng tóng de tú xíng。

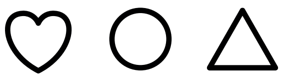
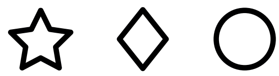
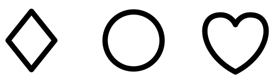
小丽、小贝、小车和小迪各有 3 个图形，每位小朋友都恰好有一个和其他小朋友相同的图形。



Which shapes does Dimitry have?

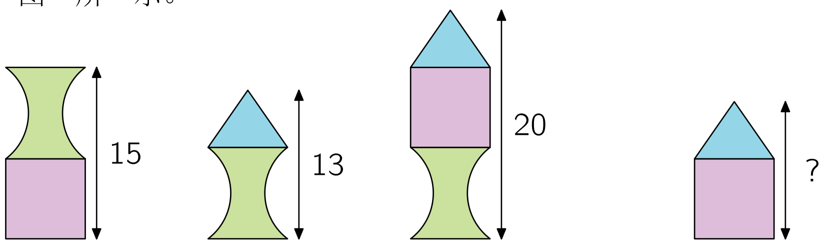
qǐng wèn xiǎo dí yǒu nǎ xiē tú xíng?

请问小迪有哪些图形?

(**A**) 
(**B**) 
(**C**) 
(**D**) 
(**E**) 

21. Zoran builds towers from three types of blocks. The heights of three of them are shown in the picture.

xiǎo lán yòng sān zhǒng bù tóng de jī mù dā le jǐ zuò tǎ, qí zhōng sān zuò tǎ de
小 兰 用 三 种 不 同 的 积 木 搭 了 几 座 塔，其 中 三 座 塔 的
gāo dù rú tú suǒ shì
高 度 如 图 所 示。

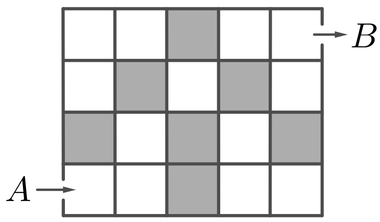


What is the height of the fourth tower?

qǐng wèn dì sì zuò tǎ de gāo dù shì duō shǎo
请 问 第 四 座 塔 的 高 度 是 多 少?

(**A**) 12 (**B**) 13 (**C**) 14 (**D**) 16 (**E**) 17

22. Zara wants to move through the grid from A to B. She can only move to the right or upwards. Each time she visits a grey box, she has to pay 1 euro. Each time she visits a white box, she has to pay 2 euros. How much would she pay for the cheapest path?



xiǎo zhā xiǎng chuān guò fāng gé, cóng A chù zǒu dào B chù。 tā zhǐ néng xiàng yòu huò
小 扎 想 穿 过 方 格，从 A 处 走 到 B 处。 她 只 能 向 右 或
xiàng shàng zǒu。 měi dāng tā zǒu dào huī sè xiǎo fāng gé shí, tā xū yào zhī fù 1
向 上 走。 每 当 她 走 到 灰 色 小 方 格 时，她 需 要 支 付 1
ōu yuán。 měi dāng tā zǒu dào bái sè xiǎo fāng gé shí, tā xū yào zhī fù 2 ōu
欧 元。 每 当 她 走 到 白 色 小 方 格 时，她 需 要 支 付 2 欧
yuán。 qǐng wèn cóng A chù zǒu dào B chù zuì shǎo xū yào huā fèi duō shǎo ōu yuán
元。 请 问 从 A 处 走 到 B 处 最 少 需 要 花 费 多 少 欧 元?

(**A**) 11 euros 11 ōu yuán 欧元 (**B**) 12 euros 12 ōu yuán 欧元 (**C**) 13 euros 13 ōu yuán 欧元

(**D**) 15 euros 15 ōu yuán 欧元 (**E**) 16 euros 16 ōu yuán 欧元

23. Julia has a list of problems to finish during May. She starts on the 1st of May.

小丽五月份需要完成一套试卷，她从五月一日开始做题。

MAY 2024

Mon	Tue	Wed	Thu	Fri	Sat	Sun
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

If she solves exactly 2 problems each day, she will finish the task on a Sunday. If she solves exactly 3 problems each day, she will finish the task on a Wednesday. How many problems are on the list?

如果她每天恰好能解决 2 道题，她将在某个星期日完成整套试卷；如果她每天恰好能解决 3 道题，她将在某个星期三完成整套试卷。请问这套试卷中一共有多少道题?

(**A**) 6 (**B**) 12 (**C**) 18 (**D**) 24 (**E**) 30

24. Andrew was throwing darts at a target. He started with 10 darts and got 2 new darts each time he hit the target. In total, Andrew threw 20 darts and then had no darts left. How many times did Andrew hit the target?

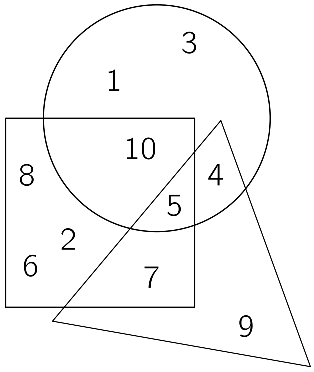
小安正在朝着靶心投掷飞镖。一开始他有 10 个飞镖，每次击中靶心后他都会获得 2 个新飞镖。小安一共投掷了 20 个飞镖，并且最后没有剩下飞镖。请问小安一共击中了多少次靶心?

(**A**) 4 (**B**) 5 (**C**) 6 (**D**) 8 (**E**) 10

Pre-Ecolier

3 points

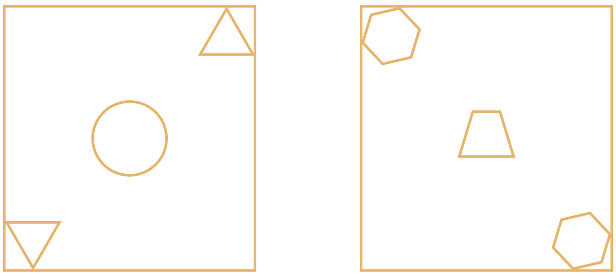
1 (China). Which number is inside the triangle, the square and the circle?

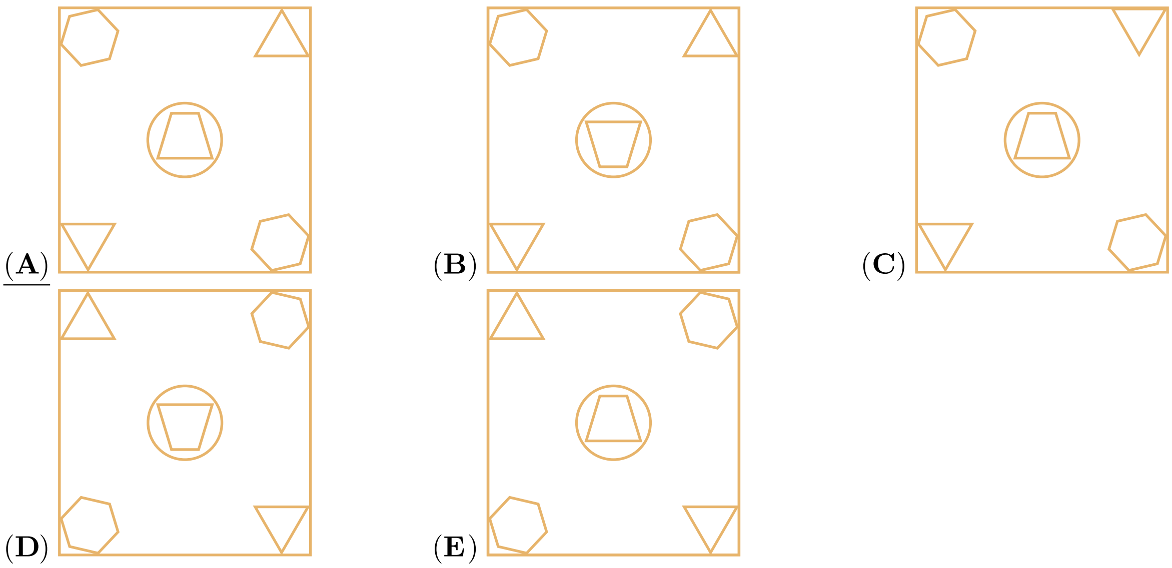


(A) 1 **(B)** 4 **(C)** 5 **(D)** 9 **(E)** 12

SOLUTION: Looking at the picture, number 5 is the only number in all three shapes.

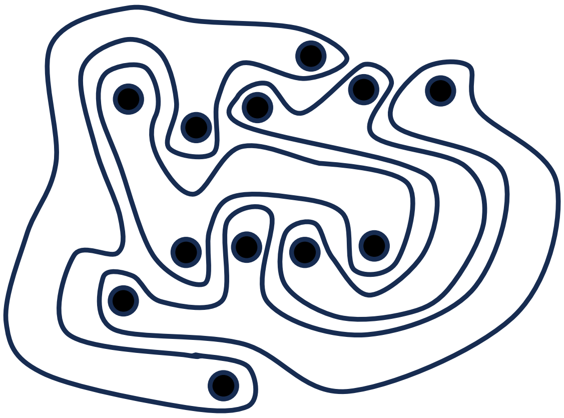
2 (China). Some shapes are printed on 2 pieces of glass. Anna places one on top of the other, without turning either piece. What does she see?





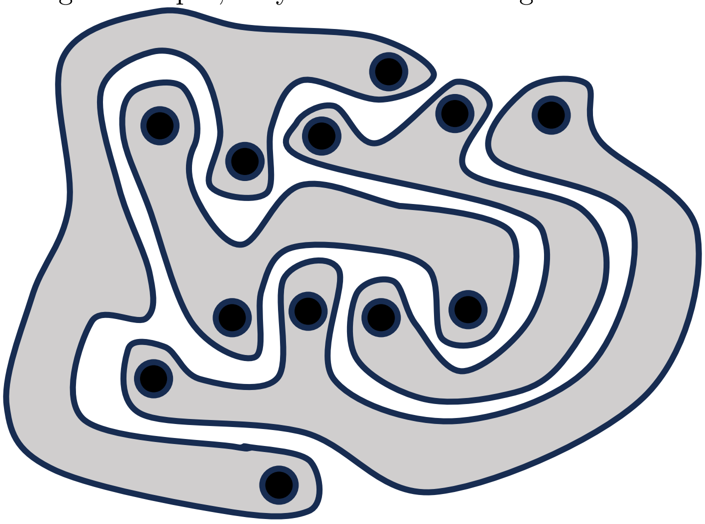
SOLUTION: When the shapes overlap, the hexagons remain in the top left and bottom right corners. Also the triangles remain in the top right and bottom left corners. This eliminates choices (D) and (E). Next we can eliminate choice (B) because the shape in the centre is inverted. Looking more closely at choice (C), we notice that the triangle in the top right corner is inverted. That leaves us with choice (A).

3 (Greece). The picture shows 4 strange shapes. How many shapes have 3 dots inside?



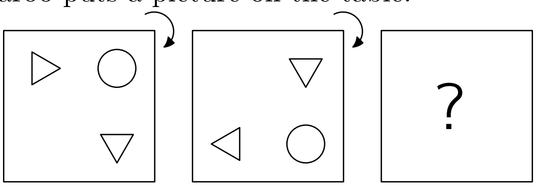
(**A**) 0 (**B**) 1 (**C**) 2 (**D**) 3 (**E**) 4

Solution: By colouring the shapes, they are easier to recognize. Then we can see that all 4 shapes

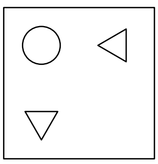
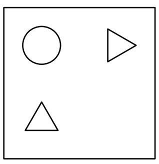
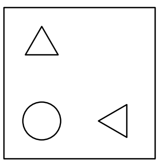
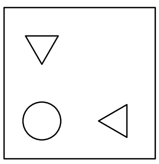
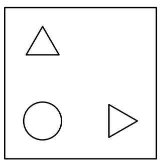


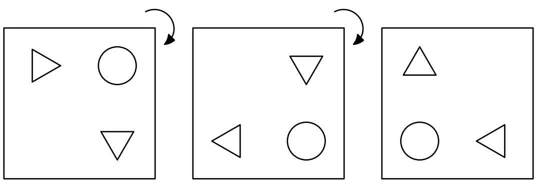
have 3 dots inside.

4 (Russia). Kevin the kangaroo puts a picture on the table.

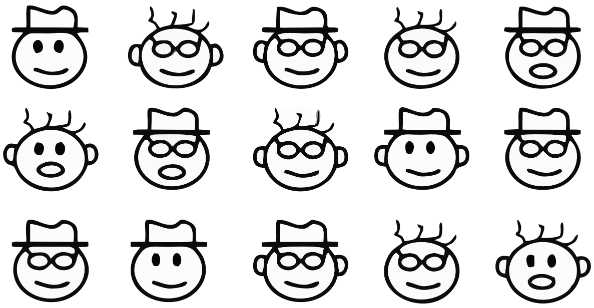


He rotates the picture through a quarter turn, as shown. He then does the same rotation again. What does Kevin see now?

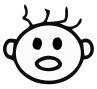
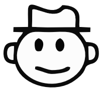
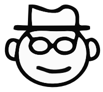
(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

Solution: 

5 (Catalonia). In the picture, there are 8 different faces.

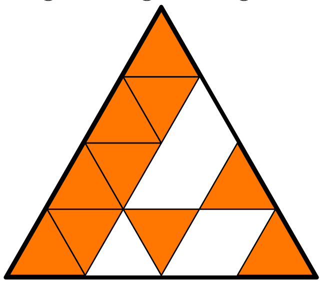


Each face appears twice, except for one. Which face appears only once?

(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

SOLUTION: After matching all the possible pairs, (C) is the only face that appears just once.

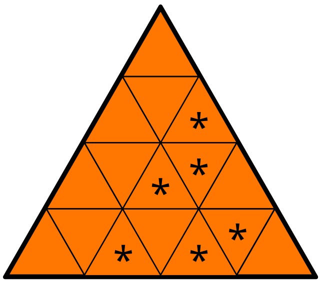
6 (Malaysia). Bruno is making this large triangle using identical small triangular tiles.



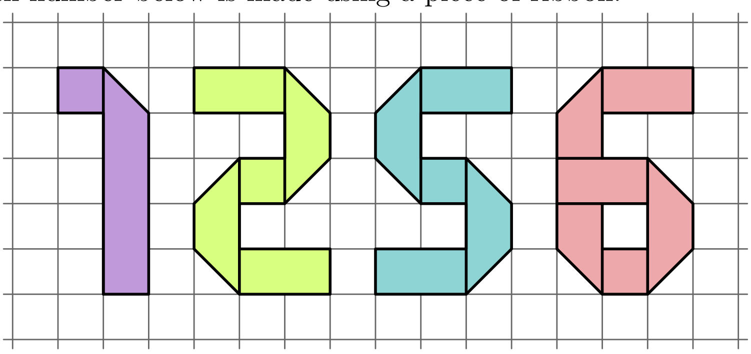
How many more tiles does Bruno need to complete the large triangle?

(**A**) 3 (**B**) 4 (**C**) 5 (**D**) 6 (**E**) 7

SOLUTION: Bruno needs six identical triangle tiles to complete the picture, as shown.



7 (Myanmar). Each number below is made using a piece of ribbon.



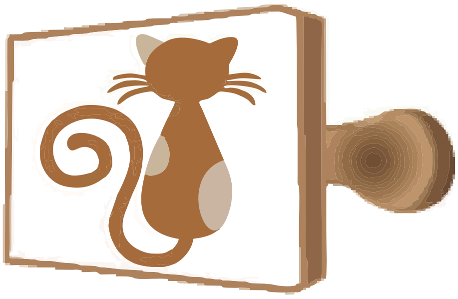
Which piece of ribbon is the longest?

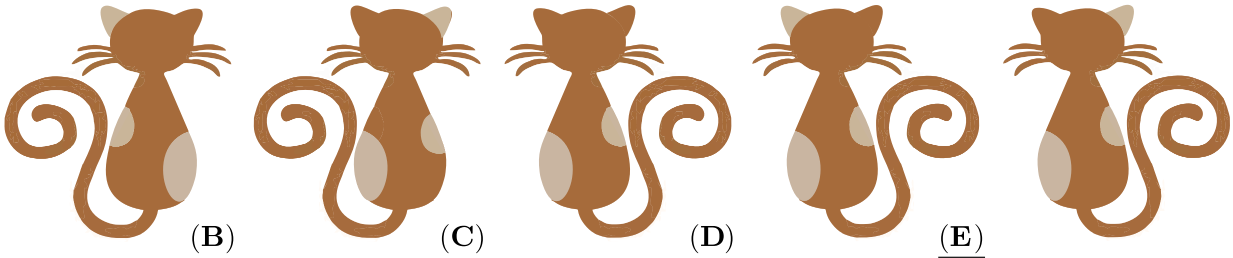
(**A**) 1 (**B**) 2 (**C**) 5 (**D**) 6

(**E**) They are all the same length.

Solution: Number '1' is the shortest one. Number '2' and '5' are mirror images, so they are the same size. Then we can see that '6' overs more squares than '5'. So '6' is the longest.

8 (Slovakia). Elena uses the stamp shown to make a picture. Which picture does she make?



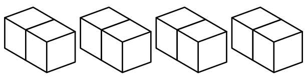


(**A**) (**B**) (**C**) (**D**) (**E**)

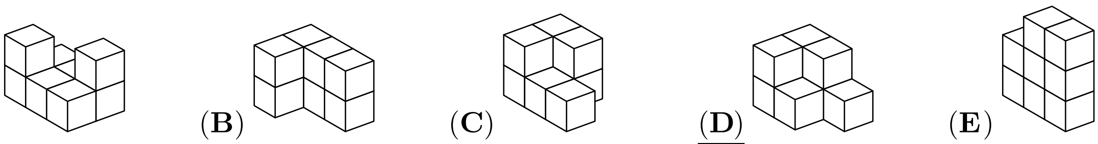
Solution: The image will be flipped when it is stamped on the paper. So (A) and (B) are not the solutions. (C) does not have different coloured ears, and (D) has coloured ears on the same side as the stamp. The correct answer is (E).

4 points

9 (Greece). A student has 4 blocks, as shown.



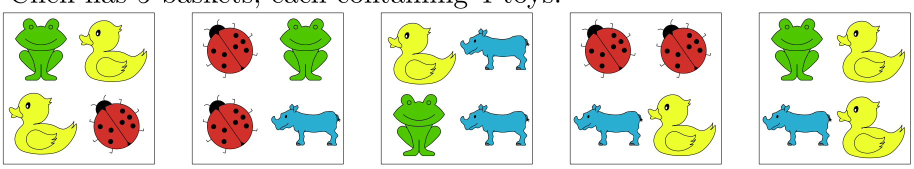
Which of the following shapes cannot be made using these 4 blocks?



(**A**) (**B**) (**C**) (**D**) (**E**)

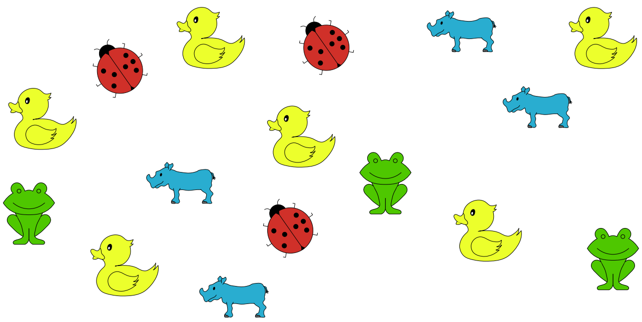
Solution: In figure D, it is not possible to put the 4 blocks together in such a way that the object could be created.

10 (Russia). Chen has 5 baskets, each containing 4 toys.



A B C D E

He dropped 4 of the baskets and the toys were mixed up.

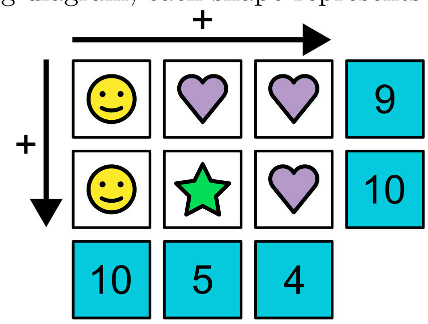


Which basket did he not drop?

(**A**) A   (**B**) B   (**C**) C   (**D**) D   (**E**) E

Solution: There are 6 ducks in the mixed picture and 6 ducks in total across the 5 boxes. So all the boxes containing ducks were dropped, and the only one that was not dropped was (B).

11 (Indonesia). In the following diagram, each shape represents a different value.

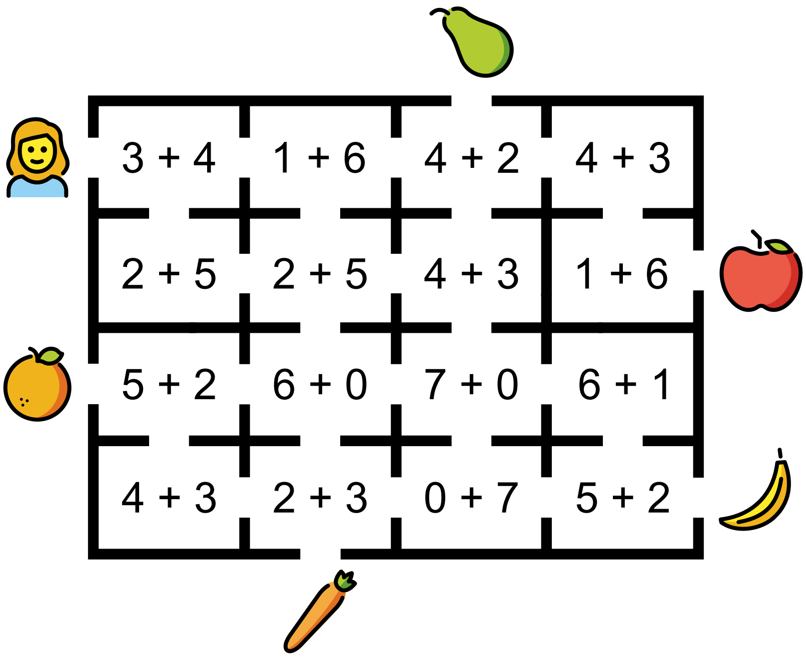


What is the value of ?

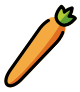
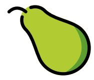
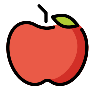
(**A**) 2   (**B**) 3   (**C**) 4   (**D**) 5   (**E**) 6

Solution: As two hearts are 4, one heart is 2. As one heart and one star are 5, one star is $5 - 2 = 3$.

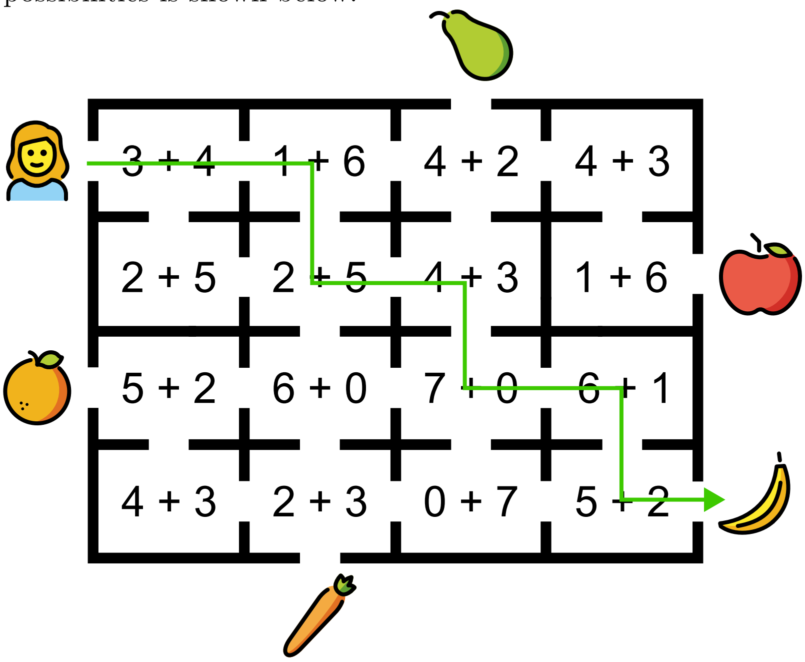
12 (Poland). Catrina wants to walk through the maze so that she visits only rooms where the answer to the sum is 7.



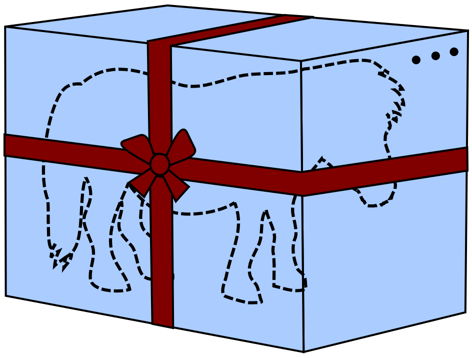
Which object can Catrina reach?

(A)  **(B)**  **(C)**  **(D)**  **(E)** 

SOLUTION: First we find the sums. Then, we walk through the maze only visiting rooms with the sum 7. There is more than one possibility, but every possible way leads to the banana. One of the possibilities is shown below.



13 (Finland). A toy pony is inside a box that is 1 metre tall, 1 metre wide and 2 metres long.

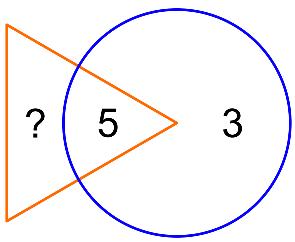


A ribbon goes around the box, as shown. The knot uses an extra 1 metre of ribbon. How long is the ribbon in total?

(A) 9 metres **(B)** 11 metres **(C)** 13 metres **(D)** 15 metres **(E)** 17 metres

SOLUTION: We need 2×2 metres for the long sides; this is 4 metres. For the tall sides, we need $2 \times 1 = 2$ metres. For the wide side, we need $4 \times 1 = 4$ metres. For the knot, we need 1 metre. In total, the ribbon is $4 + 2 + 4 + 1 = 11$ metres long.

14 (Egypt). The sum of the numbers in the triangle should be twice the sum of the numbers in the circle.

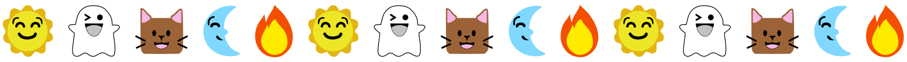


What number must replace the question mark?

(**A**) 3 (**B**) 5 (**C**) 8 (**D**) 11 (**E**) 16

SOLUTION: The sum of the numbers in the circle is 8. The sum of the numbers in the triangle must then be $2 \times 8 = 16$. So we must subtract 5 from 16 to get the answer. Doing that we have $16 - 5 = 11$.

15 (Catalonia). A line of pictures is made by repeating this pattern of 5 pictures  always in the same order.

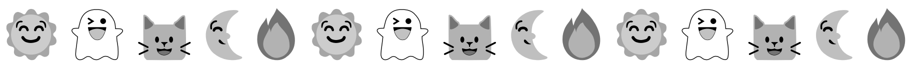
 …

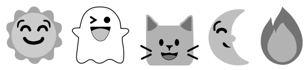
Which picture is in the 27th position in the line?

(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

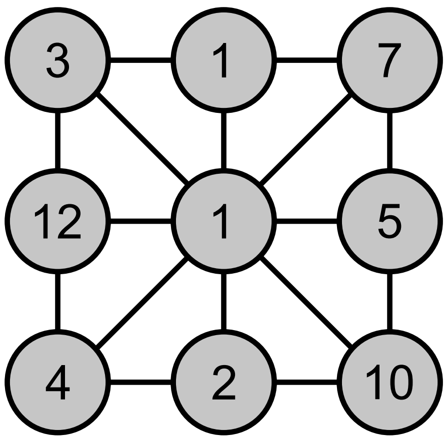
SOLUTION: The pattern repeats after every 5 pictures. So the 25th figure is a flame, the 26th is a sun, and the 27th is a ghost.

BW versions: 

 …

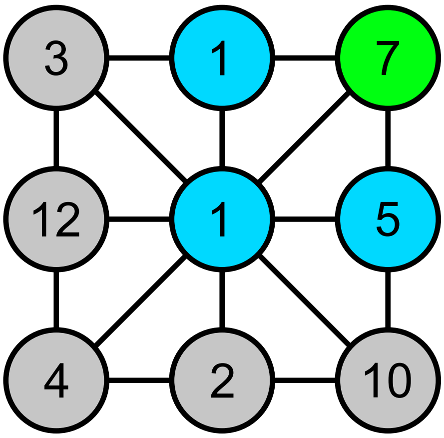


16 (Turkey). One of the numbers in the picture is equal to the sum of the numbers connected directly to it. Which number is this?



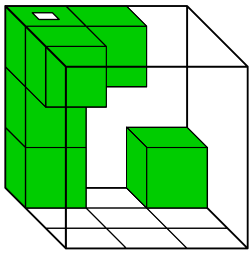
(**A**) 3 (**B**) 5 (**C**) 7 (**D**) 10 (**E**) 12

SOLUTION: If you take the number 3, you can see connecting lines to the numbers 12, 1 and 1. $12 + 1 + 1 = 14$. 14 is greater than 3. If we do this with all numbers, we see that 7 is the only number whose neighbours add up to the sum of $7 = 1 + 1 + 5$.

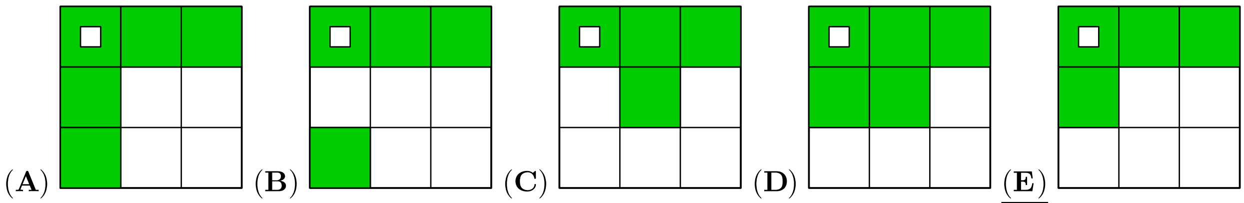


5 points

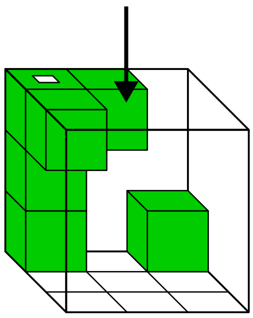
17 (Greece). Chiara has a transparent box containing 6 small cubes, as shown.



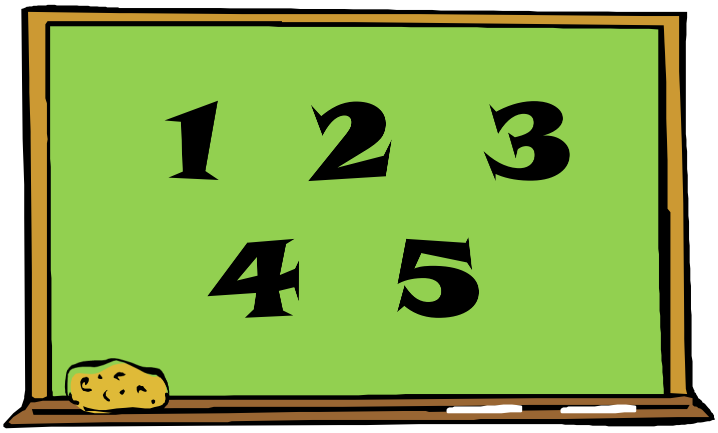
What does Chiara see if she looks at the box from above?



SOLUTION: If we look at the object from above, we see three cubes in the upper left corner. At the bottom right is a cube that we can see through the transparent glass. This cube can also be seen from above.



18 (Greece). Esteban wants to pick two numbers from the board and add them together.



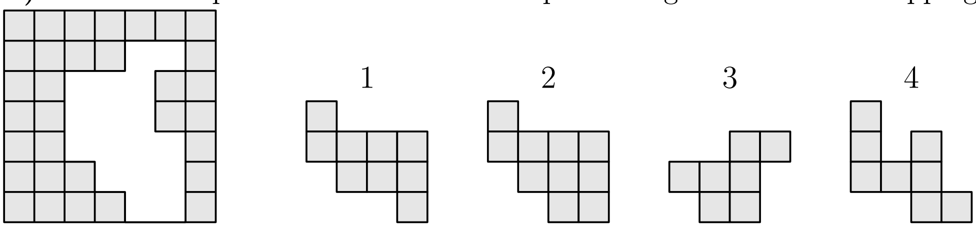
How many different results could Esteban get?

(**A**) 5 (**B**) 6 (**C**) 7 (**D**) 8 (**E**) 10

SOLUTION: The smallest possible answer he can get is $1 + 2 = 3$ and the largest is $4 + 5 = 9$. All the other numbers he can get are in between those two. In other words, all possible numbers he can get are definitely among 3, 4, 5, 6, 7, 8, 9. In fact he can get all these numbers as seen from the calculations $1 + 2 = 3$; $1 + 3 = 4$; $1 + 4 = 5$; $2 + 4 = 6$; $3 + 4 = 7$; $3 + 5 = 8$; $4 + 5 = 9$. Counting them we see that he can get 7 possible different results.

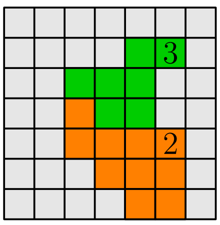
=== alternative solution === Writing out the possible sums in order: $1 + 2 = 3$; $1 + 3 = 4$; $1 + 4 = 5$; $1 + 5 = 6$; $2 + 3 = 5$; $2 + 4 = 6$; $2 + 5 = 7$; $3 + 4 = 7$; $3 + 5 = 8$; $4 + 5 = 9$. We get 10 sums overall but 3 of them have equal answers, so we can get 7 different answers.

19 (Ecuador). Which two pieces can be used to complete the grid without overlapping?



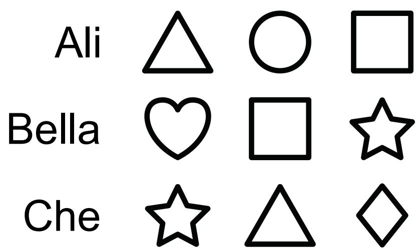
(**A**) 1 and 2 (**B**) 1 and 3 (**C**) 3 and 4 (**D**) 2 and 4 (**E**) 2 and 3

SOLUTION:

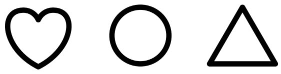


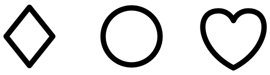
So we need piece 2 and piece 3 to complete the grid. Our solution is E.

20 (Russia). Ali, Bella, Che and Dimitry each have 3 shapes. Each child has exactly one shape in common with every other child.

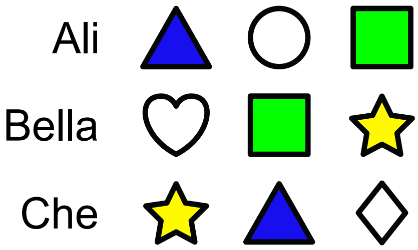


Which shapes does Dimitry have?

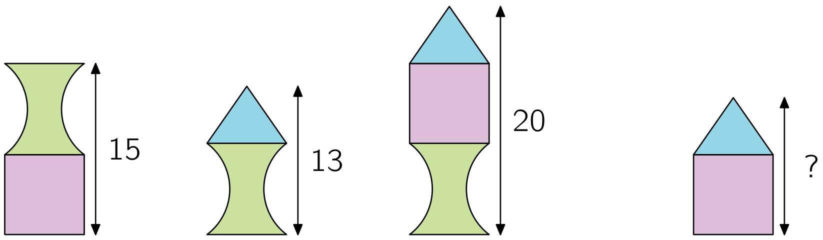
(**A**) 
(**B**) 
(**C**) 

(**D**) 
(**E**) 

SOLUTION: Ali and Bella both have a square. Bella and Che both have a star. Ali and Che both have a triangle. So Dimitry must have a circle like Ali, a heart like Bella, and a square like Che.

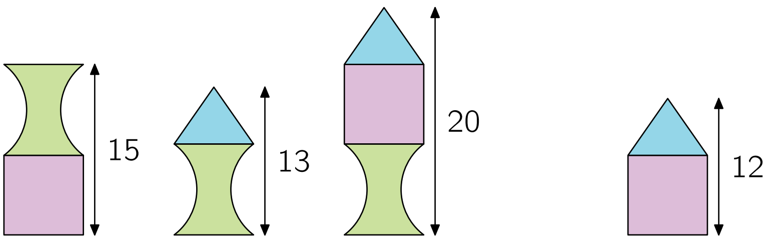


21 (Poland). Zoran builds towers from three types of blocks. The heights of three of them are shown in the picture.



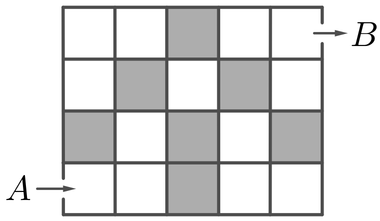
What is the height of the fourth tower?

(**A**) 12 (**B**) 13 (**C**) 14 (**D**) 16 (**E**) 17

SOLUTION: 

If we compare the first and third tower, we find that the triangle is 5 units high $(20 - 15 = 5)$. Now, if we compare the second and third tower, we find that the rectangle is 7 units high. $(20 - 13 = 7)$. The tower made from a rectangle and a triangle is therefore $12 = 7 + 5$ units high.

22 (Catalonia). Zara wants to move through the grid from A to B. She can only move to the right or upwards. Each time she visits a grey box, she has to pay 1 euro. Each time she visits a white box, she has to pay 2 euros. How much would she pay for the cheapest path?

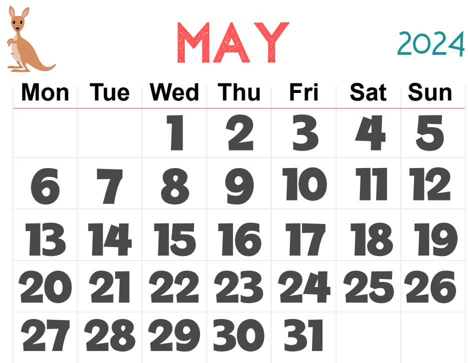


(**A**) 11 euros (**B**) 12 euros (**C**) 13 euros

(**D**) 15 euros (**E**) 16 euros

SOLUTION: Zara has to pay more for a white box than for a grey box. So she has to go through as few white boxes as possible. To get from A to B, she has to take 8 steps: 5 to the right and 3 up. We can find some paths that go through 5 white boxes, but no path that goes through 4 white boxes. So the cost of the cheapest way is $5 \times 2 + 3 \times 1 = 13$ euros.

23 (Brazil). Julia has a list of problems to finish during May. She starts on the 1st of May.



If she solves exactly 2 problems each day, she will finish the task on a Sunday. If she solves exactly 3 problems each day, she will finish the task on a Wednesday. How many problems are on the list?

(**A**) 6 (**B**) 12 (**C**) 18 (**D**) 24 (**E**) 30

SOLUTION: If Julia solves 2 problems per day and finishes on a Sunday, then there must be either 10, 24, 38 or 52 problems on the list. If Julia solves 3 problems per day and finishes on a Wednesday, then there must be either 3, 24, 45, 66 or 87 problems. The only number that these two lists have in common is 24.

24 (Belarus). Andrew was throwing darts at a target. He started with 10 darts and got 2 new darts each time he hit the target. In total, Andrew threw 20 darts and then had no darts left. How many times did Andrew hit the target?

(**A**) 4 (**B**) 5 (**C**) 6 (**D**) 8 (**E**) 10

SOLUTION: For each hit, Andrew gets two more darts. He threw 20 in total, and so must have got $20 - 10 = 10$ more than he had at the beginning. He would have got 10 extra darts by hitting the target $10 \div 2 = 5$ times.